enough air. What is checked here is essentially the compressibility of air. The tyre should be less compressible for its easy rolling



In fact the rear tyre is less compressible than front tyre for a smooth ride.

EXAMPLE 7.3

A metallic cube of side 100 cm is subjected to a uniform force acting normal to the whole surface of the cube. The pressure is 10^6 pascal. If the volume changes by 1.5×10^{-5} m^3, calculate the bulk modulus of the material.

Solution

By definition, $\mathrm{K} = \dfrac{\dfrac{F}{A}}{\dfrac{\Delta V}{V}} = P\dfrac{V}{\Delta V}$

$$\mathrm{K} = \frac{10^6 \times 1}{1.5 \times 10^{-5}} = 6.67 \times 10^{10}\ \mathrm{N\,m^{-2}}$$

Rigidity modulus or shear modulus:
The rigidity modulus is defined as the ratio of the shearing stress to the shearing strain.

$$\eta_R = \frac{\text{Shearing stress}}{\text{Angle of shear or shearing strain}}$$

The shearing stress is

$$\sigma_s = \frac{\text{Tangential force}}{\text{Area over which it is applied}} = \frac{F_t}{\Delta A}$$

The angle of shear or shearing strain

$$\varepsilon_s = \frac{x}{h} = \theta$$

Therefore, Rigidity modulus is

$$\eta_R = \frac{\sigma_s}{\varepsilon_s} = \frac{\dfrac{F_t}{\Delta A}}{\dfrac{x}{h}} = \frac{F_t}{\Delta A \theta} \qquad (7.9)$$

Further, the equation (7.9) implies that a material can be easily twisted if it has small value of rigidity modulus. For example, consider a wire twisted through an angle θ, a restoring torque (τ) developed is

$$\tau \propto \theta$$

This means that for a larger torque, wire will twist by a larger amount (angle of shear θ is large). Since the rigidity modulus is inversely proportional to angle of shear, the modulus of rigidity is small.

For the best understanding, the elastic coefficients of some of the important materials are listed in Table 7.1.

Table 7.1 Elastic coefficient of some materials

Material	Young's modulus (Y) (10^{10}N m^{-2})	Bulk modulus (K) (10^{10} N m^{-2})	Shear modulus (η_R) (10^{10} N m^{-2})
Steel	20.0	15.8	8.0
Aluminium	7.0	7.0	2.5
Copper	12.0	12.0	4.0
Iron	19.0	10	6.4
Glass	7.0	3.6	2.6

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